

Hyphenation: English

Electroencephalography (EEG) is a method to record an electrogram of the spontaneous electrical activity of the brain. The bio signals detected by EEG have been shown to represent the postsynaptic potentials of pyramidal neurons in the neocortex and allocortex. It is typically non-invasive, with the EEG electrodes placed along the scalp (commonly called "scalp EEG") using the International 10–20 system, or variations of it. Electrocorticography, involving surgical placement of electrodes, is sometimes called "intracranial EEG". EEG is widely used both as a clinical diagnostic tool, particularly in epilepsy, and as a research tool in neuroscience. Clinical interpretation of EEG recordings is performed by visual inspection of the tracing, which is the standard method.

Quantitative EEG analysis may be used as an adjunct in specific clinical settings. Visual interpretation of EEG is subject to inter-rater and intra-rater variability. Voltage fluctuations measured by the EEG bio amplifier and electrodes allow the evaluation of normal brain activity. As the electrical activity monitored by EEG originates in neurons in the underlying brain tissue, the recordings made by the electrodes on the surface of the scalp vary in accordance with their orientation and distance to the source of the activity. Furthermore, the value recorded is distorted by intermediary tissues and bones, which act in a manner akin to resistors and capacitors in an electrical circuit.

This means that not all neurons will contribute equally to an EEG signal, with an EEG predominately reflecting the activity of cortical neurons near the electrodes on the scalp. Deep structures within the brain further away from the electrodes will not contribute directly to an EEG; these include the base of the cortical gyrus, medial walls of the major lobes, hippocampus, thalamus, and brain stem. A healthy human EEG will show certain patterns of activity that correlate with how awake a person is. The range of frequencies one observes are between 1 and 30 Hz, and amplitudes will vary between 20 and 100 μV . The observed frequencies are subdivided into various groups: alpha (8–13 Hz), beta (13–30 Hz), delta (0.5–4 Hz), and theta (4–7 Hz).

Alpha waves are observed when a person is in a state of relaxed wakefulness and are mostly prominent over the parietal and occipital sites. During intense mental activity, beta waves are more prominent in frontal areas as well as other regions. If a relaxed person is told to open their eyes, one observes alpha activity decreasing and an increase in beta activity.

Theta and delta waves are not generally seen in wakefulness – if they are, it is a sign of brain dysfunction. EEG can detect abnormal electrical discharges such as sharp waves, spikes, or spike-and-wave complexes, as observable in people with epilepsy; thus, it is often used to inform medical diagnosis.

Psychopharmacology (from Greek ψῡχή, psūkhē, 'breath, life, soul'; φάρμακον, pharmakon, 'drug'; and -λογία, -logia) is the scientific study of the effects drugs have on mood, sensation, thinking, behavior, judgment and evaluation, and memory. It is distinguished from neuropsychopharmacology, which emphasizes the correlation between drug-induced changes in the functioning of cells in the nervous system and changes in consciousness and behavior. The field of psychopharmacology studies a wide range of substances with various types of psychoactive properties, focusing primarily on the chemical interactions with the brain. The term "psychopharmacology" was likely coined by David Macht in 1920.

Psychoactive drugs interact with particular target sites or receptors found in the nervous system to induce widespread changes in physiological or psychological functions. The specific interaction between drugs and their receptors is referred to as "drug action", and the widespread changes in physiological or psychological function is referred to as "drug effect". These drugs may originate from natural sources such as plants and animals, or from artificial sources such as chemical synthesis in the laboratory.

Not often mentioned or included in the field of psychopharmacology today are psychoactive substances not identified as useful in modern mental health settings or references. These substances are naturally occurring, but nonetheless psychoactive, and are compounds identified through the work of ethnobotanists and ethnomycologists (and others who study the native use of naturally occurring psychoactive drugs). However, although these substances have been used throughout history by various cultures, and have a profound effect on mentality and brain function, they have not always attained the degree of scrupulous evaluation that lab-made compounds have.

Nevertheless, some, such as psilocybin and mescaline, have provided a basis of study for the compounds that are used and examined in the field today. Hunter-gatherer societies tended to favor hallucinogens, and today their use can still be observed in many surviving tribal cultures. The exact

drug used depends on what the particular ecosystem a given tribe lives in can support, and are typically found growing wild. Such drugs include various psychoactive mushrooms containing psilocybin or muscimol and cacti containing mescaline and other chemicals, along with myriad other plants containing psychoactive chemicals. These societies generally attach spiritual significance to such drug use, and often incorporate it into their religious practices.

With the dawn of the Neolithic and the proliferation of agriculture, new psychoactives came into use as a natural by-product of farming. Among them were opium, cannabis, and alcohol derived from the fermentation of cereals and fruits.